

6. Write short notes on any **three** of the following:

3×5=15

- (a) Restriction enzymes
- (b) Chimeric DNA
- (c) Knockout mice
- (d) Cloning Vectors

[This question paper contains 4 printed pages]

Your Roll No.

: 19.05.2026(M)

Sl. No. of Q. Paper

: 1738

Unique Paper Code

: 2232013601

Name of the Paper

: Animal Biotechnology
NEP-UGCF

Name of the Course

: B.Sc.(H) Zoology

Semester

: VI

Time : 2 Hours

Maximum Marks : 60

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) Attempt **four** questions including Question No. 1 which is compulsory.

1. (a) Define (any **three**):

1×3=3

- (i) Insertional inactivation
- (ii) In vivo gene therapy
- (iii) Transgenesis
- (iv) Expression vector

- (b) Distinguish between (**any three**): $2 \times 3 = 6$
- Blunt and Cohesive ends
 - Isochizomers and Isocaudomers
 - Dolly and Polly
 - M13 phage and lambda phage
- (c) Expand the following: $0.5 \times 4 = 2$
- OLA-PCR
 - SCNT
 - TALEN
 - ASO
- (d) Give the important contributions of the following scientists: $1 \times 4 = 4$
- D. T. Burke and G. F. Carle
 - H. W. Boyer and S. N. Cohen
 - Ian Wilmut
 - Werner Arber
2. (a) Describe the method of construction of cDNA library with the help of suitable diagram 8

- (b) Explain Colony hybridization method for screening of genomic libraries. 7
3. (a) Describe microinjection method for production of transgenic animals. 8
- (b) Explain application of transgenic animals in production of pharmaceuticals. 7
4. (a) Describe the procedure of production of Humulin by recombinant DNA technology 8
- (b) Explain molecular diagnostic techniques for detection of Sickle cell anemia. 7
5. (a) Explain the CRISPR/Cas9 system and its applications in gene editing 8
- (b) Explain SCID. Describe method used for its gene therapy 7